

```
===== NUMPY ANALYZER =====
```

```
===== MAIN MENU =====
```

1. Create Array
2. Indexing & Slicing
3. Mathematical Operations
4. Combine And Split Arrays
5. Search,Sort and Filter Array
6. Aggregate & Statistics
7. Exit

Enter your choice: 1

Array Creation:

1. 1D Array
2. 2D Array
3. 3D Array

Enter your choice: 2

```
Enter your choice: 2
Enter rows: 2
Enter columns: 3
Enter 6 elements: 10 20 30 40 50 60
```

```
Array Created Successfully:
[[10 20 30]
 [40 50 60]]
```

```
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Enter your choice: 2

1. Indexing
2. Slicing

Enter choice: 1

Enter 2 index values: 1 2

Element = 60

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Enter your choice: 2

1. Indexing
2. Slicing

Enter choice: 2

Dimension 0 slice (start:end): 0:2

Dimension 1 slice (start:end): 1:3

Sliced Array:

```
[[20 30]
 [50 60]]
```

===== MAIN MENU =====

1. Create Array
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Mathematical Operations:

1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter choice: 1

Enter 6 elements: 1 2 3 4 5 6

Result:

```
[[11 22 33]
 [44 55 66]]
```

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1. Create Array
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6. Aggregate & Statistics

7. Exit

Enter your choice: 3

Mathematical Operations:

1. Addition

2. Subtraction

3. Multiplication

4. Division

Enter choice: 2

Enter 6 elements: 1 1 1 1 1 1

Result:

[[9 19 29]

[39 49 59]]

===== MAIN MENU =====

1. Create Array

Enter your choice: 3

Mathematical Operations:

1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter choice: 3

Enter 6 elements: 2 2 2 2 2 2

Result:

```
[[ 20  40  60]
 [ 80 100 120]]
```

Mathematical Operations:

1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter choice: 4

Enter 6 elements: 2 2 2 2 2 2

Result:

```
[[ 5. 10. 15.]  
 [20. 25. 30.]]
```


6. Aggregate & Statistics

7. Exit

Enter your choice: 4

1. Combine Arrays

2. Split Array

Enter choice: 1

Enter 6 elements: 1 2 3 4 5 6

Combined Array:

[[10 20 30]

[40 50 60]

[1 2 3]

[4 5 6]]

===== MAIN MENU =====

1. Create Array

2. Indexing & Slicing

1. Create Array
2. Indexing & Slicing
3. Mathematical Operations
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7. Exit

Enter your choice: 4

1. Combine Arrays
2. Split Array

Enter choice: 2

Enter number of splits: 2

Split Arrays:

[[10 20 30]]

[[40 50 60]]

```
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```
Enter your choice: 5
```

1. Search Value
2. Sort Array
3. Filter Values

```
Enter choice: 1
```

```
Enter value to search: 30
```

```
Found at indices: (array([0]), array([2]))
```

Enter your choice: 5

1. Search Value
2. Sort Array
3. Filter Values

Enter choice: 2

Ascending (a) or Descending (d): a

[10 20 30 40 50 60]

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Enter your choice: 6

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 1

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 1

Sum = 210

===== MAIN MENU =====

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1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 2

Mean = 35.0

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6. Aggregate & Statistics

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 3

Median = 35.0

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 4

Std Dev = 17.07825127659933

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 5

Variance = 291.666666666667

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 6

Min = 10 Max = 60

1. Sum
2. Mean
3. Median
4. Standard Deviation
5. Variance
6. Min & Max
7. Percentile

Enter choice: 7

Enter percentile: 90

90th Percentile = 55.0

```
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```
Enter your choice: 7
```

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